

BESS — 40ft 3.5 MWh containerised

Revision Rev A · Fractional Forge · In build

Forge project pack

Project name	BESS — 40ft 3.5 MWh containerised
Revision	Rev Rev A
Project created	25 Apr 2026, 05:42
Brief locked	25 Apr 2026, 18:30
Shipped	Not shipped yet
Document generated	4 May 2026, 13:31

RED

Feasibility — blockers found

- Design does not fit the declared envelope.
- Estimated unit cost is 2.8× the brief ceiling.



BRIEF INFEASIBLE — DO NOT PROCEED TO PROCUREMENT

The design as currently briefed cannot be built within the declared envelope, mass, or transport constraints. Modules, bill of materials, cost waterfall, reconciliation, risks register and supplier shortlist are intentionally omitted from this document. Resolve the blockers listed on the Feasibility Exception page before any module decomposition or supplier engagement.

TOTALS AT A GLANCE

MODULES 9	KEY PARTS 72	BOM ROWS 18	SUPPLIERS 5
FAILURE MODES 37	OPEN QUESTIONS 27	STANDARDS 11	REVIEWS 3
UNIT COST Not computed — design infeasible	CEILING	HEADROOM	

Contents

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Feasibility

Before this report was assembled, the design was checked against the brief constraints. The verdict below is computed deterministically from the sizing solver, bill of materials, and cost waterfall — not from language-model opinion.

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WHAT WE ARE BUILDING

A 40-foot containerised battery energy storage system delivering 3.5 MWh and 1.5 MW for behind-the-meter commercial and industrial sites in the United Kingdom. Lithium iron phosphate cells, integrated battery management, bi-directional power conversion with grid-forming capability, liquid-cooled thermal management, and certified fire detection and suppression. Target installed capital cost under 180,000 pounds. Must ship complete on a flatbed lorry, install on a prepared concrete pad, and commission inside two days. Compliant with United Kingdom G99 grid code and the latest National Fire Chiefs Council guidance for lithium battery storage.

MISSION

Deliver a factory-integrated, containerised lithium iron phosphate battery energy storage system that enables UK commercial and industrial sites to optimise energy costs and support grid stability behind the meter, without specialist on-site build work.

USE CASE

A prepared-pad, two-day-commission containerised BESS that ships complete on a flatbed lorry and connects behind the meter to reduce peak demand charges and provide grid services for UK C&I sites.

TARGET CUSTOMERS

UK commercial and industrial site operators, energy-intensive manufacturers, logistics and distribution centres, property developers with large commercial tenants

WHY NOW

UK grid code G99 now mandates grid-forming capability for new behind-the-meter storage connections; NFCC guidance on lithium battery storage has matured into a de-facto planning requirement; LFP cell costs have fallen sufficiently to make containerised C&I BESS commercially viable at scale; DNOs are actively incentivising demand-side flexibility.

Constraints declared

Unit cost ceiling	—
Max mass	40,000.00 kg

SECTION JUDGEMENT — COMPOSITE SCORE: 6.8/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	8/10	Constraint data aligns well with realistic physical limits for a 40ft container.	<i>I, Gemini, will add source verification checks to ensure physical limits match real-world norms.</i>
Visual Clarity	5/10	Heavy text blocks make scanning constraints difficult without visual separation.	<i>I, Gemini, will implement a structured Key-Value pair format. Constraints should be a bulleted checklist rather than paragraphs.</i>
Linguistic Precision	7/10	Terminology is industry-standard, though occasionally verbose.	<i>I, Gemini, will reduce verbosity by extracting common nouns into a glossary.</i>
Logical Coherence	8/10	Mission, use case, and rationale follow a solid deductive flow.	<i>I, Gemini, will enforce a stricter mapping between mission statements and technical requirements.</i>
Actionability	6/10	Provides targets but lacks explicit constraint checklists for engineering.	<i>I, Gemini, will parse targets into measurable KPIs with pass/fail thresholds.</i>

What is good:

Narrative strength is excellent. The constraints provide solid boundary definitions.

What is bad:

Walls of text create cognitive friction. The layout does not visually separate the 'Mission' from technical 'Constraints'.

2. Regulatory posture

ADB

Per-standard compliance matrix — applicability, design impact, evidence required, status, owner, and the next concrete gap action.

5 of 5 regulatory entries are unverified extractions.

These entries were populated by an automated extraction pass and should be cross-checked against the original standard text before any procurement, certification, or design-freeze decision.

ENA G99

NOT STARTED

Electrical lead

Unverified

Requirements for the Connection of Generation Equipment in Parallel with Public Distribution Networks

Mandatory UK grid code for behind-the-meter storage connecting in parallel with the distribution network; grid-forming capability required.

Applicability: Applicable to the 1.5MW grid-scale inverter.

Design impact: Requires SiC or advanced IGBT bi-directional inverter.

Evidence required: Type test certificate or individual commissioning test.

Next action: Verify SMA inverter compliance with latest G99 amendment.

BS EN IEC 62619

NOT STARTED

Battery safety lead

Unverified

Secondary Lithium Cells and Batteries — Safety Requirements for Use in Industrial Applications

Cell and module-level safety standard required by UL 9540 for LFP battery components

Applicability: UK statutory + EU — every cell + module supplied into the BESS must meet BS EN IEC 62619 testing for industrial deployment.

Design impact: Cell vendor selection limited to those with current 62619 certificates (CATL/EVE/Sunwoda); module-level abuse testing at supplier.

Evidence required: BS EN IEC 62619 test report from accredited lab (TÜV / DEKRA / Intertek) per cell SKU.

Next action: Add 62619 cert demand to the cell-supplier RFQ.

BS EN IEC 63056

NOT STARTED

Battery safety lead

Unverified

Safety requirements for secondary lithium cells and batteries for use in electrical energy storage systems

System-level safety certification required for the complete BESS installation.

Applicability: UK statutory + EU — system-level safety standard for stationary BESS.

Design impact: BMS architecture, contactor + fuse coordination, thermal runaway propagation barriers, enclosure venting design.

Evidence required: BS EN IEC 63056 test report from accredited lab; system-level Declaration of Conformity.

Next action: Book accredited-lab campaign 6+ months before commissioning.

UL 9540A

NOT STARTED

Battery safety lead

Unverified

Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems

Thermal-runaway propagation test recommended to demonstrate fire safety in support of NFCC guidance compliance.

Applicability: International (US-origin) — NFCC + UK fire authorities increasingly demand UL 9540A test data.

Design impact: Drives module-level barrier specification (mica/aerogel placement); enclosure vent area sizing; deflagration vent panel design.

Evidence required: UL 9540A test report at cell, module, unit AND installation levels.

Next action: Source cells whose vendor already has unit-level UL 9540A.

Standard for the Installation of Stationary Energy Storage Systems

Governs minimum fire separation distances, deflagration venting, and suppression requirements.

Applicability: International (US-origin) but adopted as a de-facto reference by UK NFCC.

Design impact: Fire separation: 3 m to property lines; deflagration vent area calc per NFPA 68; gas-suppression agent (Novec 1230) sizing.

Evidence required: Fire safety strategy referencing NFPA 855 clauses; vent-area calc.

Next action: Engage a fire-safety consultant 3 months before planning submission.

SECTION JUDGEMENT — COMPOSITE SCORE: 7.4/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	9/10	Citations of G99, NFPA 855, and UL 9540A are perfectly aligned with UK battery storage standards.	<i>I, Gemini, will cross-reference regional codes via an external API to ensure they haven't been superseded.</i>
Visual Clarity	5/10	Tables are prone to breaking across pages, cutting vital compliance information in half.	<i>I, Gemini, will ensure strict wrap={false} on table row <View> components so they break cleanly to the next page.</i>
Linguistic Precision	8/10	Action items ('Verify SMA inverter compliance') are clearly articulated.	<i>I, Gemini, will standardise action item verbs to form an imperative engineering backlog.</i>
Logical Coherence	8/10	Categorisation by code, summary, impact, and action logically flows from risk to resolution.	<i>I, Gemini, will group these codes by geographical applicability for better coherence.</i>
Actionability	7/10	Gap actions provide usable tasks, but are buried in table columns.	<i>I, Gemini, will extract gap actions into a stand-alone checklist at the end of the regulatory section.</i>

- What is good:

The compliance matrix structure is solid, and the unverified extraction warnings add critical trust.
- What is bad:

React-PDF struggles with large tables across page breaks. Rows clip in half.

Container Enclosure & Structure

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Modified 40-ft ISO high-cube steel container providing the weatherproof, fire-rated, structural housing for all subsystems with service aisle, doors, roof penetrations, and ISO corner castings.

WHY IT MATTERS

The container is the primary structural, environmental, and fire-safety boundary of the entire product — it carries every subsystem load, defines IP/weather rating, provides the NFPA 855 fire compartment, and is the single earth reference for the whole system.

DESCRIPTION

The enclosure is a new-build (rather than second-hand refurbished) 40-ft ISO 1AAA high-cube steel container, modified at the fabricator to suit BESS duty: reinforced undercarriage for ~3,000 kg/m^2 floor loading, pre-cut and framed roof penetrations for the ceiling HVAC unit and deflagration vents, two opposing personnel doors with panic hardware, and an internal fire-rated rockwool/steel sandwich lining giving a 1-hour rating per NFPA 855 Chapter 9.

KEY PARTS (7)

- Modified 40-ft ISO high-cube container shell, external 12,192 × 2,438 × 2,896 mm
- Reinforced steel deck floor replacing standard 28 mm plywood
- Internal thermal/fire-rated lining: 50–75 mm rockwool with 1-hour fire-rated inner steel sheet
- Personnel service door 900 × 2,100 mm single-leaf outward-opening with crash/panic bar
- Roof penetration frames for HVAC unit (~2,000 × 1,000 mm cut-out)
- Single-point earth bonding bar (copper, 40 × 5 mm)
- Fire-rated sealed cable transit penetrations (Roxtec-type, 2-hour rated)

FAILURE MODES (4)

- Floor deflection or weld cracking under concentrated battery rack loads
- Corrosion perforation of Corten walls or roof around HVAC and cable penetrations
- Fire-rated lining degradation or seam failure compromising the 1-hour compartment rating
- Loss of earth bonding continuity at rack/PCS interfaces causing touch-voltage hazard

OPEN QUESTIONS (3)

- Whether deflagration venting via roof panels or forced mechanical ventilation is preferred
- Final floor-loading spec: is 3,000 kg/m^2 sufficient
- Whether the step-up transformer is internal or external

Estimated cost £19,050

SECTION JUDGEMENT — COMPOSITE SCORE: 8.2/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.	<i>I, Gemini, will cross-check mass totals automatically against transport limits.</i>
Visual Clarity	6/10	Good flow, but images with extreme aspect ratios create jarring layout shifts.	<i>I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.</i>
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.	<i>I, Gemini, will apply a readability filter to ensure consistent noun usage.</i>
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.	<i>I, Gemini, will map failure modes to the specific components they affect.</i>
Actionability	8/10	Reviews and risk matrices attached to modules provide direct mitigations.	<i>I, Gemini, will export risk mitigations to an external issue tracker.</i>

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images push text down awkwardly if the aspect ratio is extreme.

2400.0 kg / budget 2500.0 kg · 16 wk lead · Specialist judgement

Port-side LFP Battery Rack Array

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Houses half of the LFP battery modules along one long sidewall, providing DC energy storage with integrated rack-level BMS.

WHY IT MATTERS

This module represents half of the product's stored energy and therefore half of the project's bill-of-materials cost; its cell chemistry, module format, and rack architecture drive the system's round-trip efficiency, cycle life, warranty exposure, and UL 9540A fire-propagation behaviour.

DESCRIPTION

The port-side battery rack array comprises two free-standing steel-framed racks arranged along the port long sidewall of the container, each housing 10-12 LFP prismatic-cell modules stacked vertically and interconnected by copper busbars to form a ~500 kWh / ~1500 V DC nominal string. Each rack integrates a rack-level BMU monitoring per-cell voltage and per-module temperature, a main DC contactor with pre-charge resistor and NH-style fuse for string isolation, and an insulation-monitoring tap.

KEY PARTS (7)

- 2x LFP battery racks, 600 x 1200 x 2200 mm
- 20-24x LFP prismatic-cell modules (10-12 per rack)
- Rack-level BMU (Battery Management Unit) per rack
- Rack main DC contactor and pre-charge circuit, 1500 V DC / 400 A rated
- Intra-rack DC busbars, tin-plated copper, 30 x 10 mm cross-section
- Rack top air plenum with filtered supply-air inlet
- Seismic bracing brackets, 5 mm steel

FAILURE MODES (4)

- Thermal runaway of a single cell propagating to adjacent modules
- Cell imbalance and accelerated capacity fade from BMU sensor drift
- Main DC contactor weld-closed on fault current
- Busbar connection loosening under vibration/thermal cycling

OPEN QUESTIONS (3)

- Final cell/module vendor selection (CATL, EVE, Gotion)
- Whether rack-level liquid cooling plates are required instead of forced-air
- DC string voltage target (1000 V vs 1500 V) pending final PCS selection

Estimated cost £76,000

SECTION JUDGEMENT — COMPOSITE SCORE: 8.2/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.	<i>I, Gemini, will cross-check mass totals automatically against transport limits.</i>
Visual Clarity	6/10	Good flow, but images with extreme aspect ratios create jarring layout shifts.	<i>I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.</i>
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.	<i>I, Gemini, will apply a readability filter to ensure consistent noun usage.</i>
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.	<i>I, Gemini, will map failure modes to the specific components they affect.</i>
Actionability	8/10	Reviews and risk matrices attached to modules provide direct mitigations.	<i>I, Gemini, will export risk mitigations to an external issue tracker.</i>

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3.3 · Starboard-side LFP Battery Rack Array

ADB

2400.0 kg / budget 2500.0 kg · 16 wk lead · Specialist judgement · mirrors Port-side LFP Battery Rack Array

Starboard-side LFP Battery Rack Array

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Houses the mirrored half of the LFP battery modules along the opposite long sidewall, providing DC energy storage with integrated rack-level BMS.

WHY IT MATTERS

Together with the port-side array, this module stores 100% of the system's usable energy; its reliability, thermal uniformity, and safe failure behaviour directly determine round-trip efficiency, warranty cycle life, and most critically the probability of a thermal runaway event.

DESCRIPTION

The starboard-side battery rack array is the mirrored counterpart to the port-side array, comprising 2 floor-standing LFP battery racks arranged along the starboard long sidewall of the container, separated from the central service aisle by ~1,000 mm clear working space.

KEY PARTS (7)

- 2x LFP battery racks, 600 x 1200 x 2200 mm
- 20-24x LFP prismatic-cell modules (10-12 per rack)
- Rack-level BMU (Battery Management Unit) per rack
- Rack main DC contactor and pre-charge circuit, 1500 V DC / 400 A rated
- Intra-rack DC busbars, tin-plated copper, 30 x 10 mm cross-section
- Rack top air plenum with filtered supply-air inlet
- Seismic bracing brackets, 5 mm steel

FAILURE MODES (4)

- Thermal runaway of a single cell propagating to adjacent modules
- Cell imbalance and accelerated capacity fade from BMU sensor drift
- Main DC contactor weld-closed on fault current
- Busbar connection loosening under vibration/thermal cycling

OPEN QUESTIONS (3)

- Final cell/module vendor selection (CATL, EVE, Gotion)
- Whether rack-level liquid cooling plates are required instead of forced-air
- DC string voltage target (1000 V vs 1500 V) pending final PCS selection

Estimated cost £76,100

SECTION JUDGEMENT — COMPOSITE SCORE: 8.2/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.	<i>I, Gemini, will cross-check mass totals automatically against transport limits.</i>
Visual Clarity	6/10	Good flow, but images with extreme aspect ratios create jarring layout shifts.	<i>I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.</i>
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Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.	<i>I, Gemini, will map failure modes to the specific components they affect.</i>
Actionability	8/10	Reviews and risk matrices attached to modules provide direct mitigations.	<i>I, Gemini, will export risk mitigations to an external issue tracker.</i>

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3.4 · DC Combiner & Distribution Panel

ADB

180.0 kg / budget 200.0 kg · 10 wk lead · Specialist judgement

DC Combiner & Distribution Panel

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Aggregates, fuses, and protects the DC outputs of all battery racks and routes combined DC to the PCS.

WHY IT MATTERS

The DC combiner is the single aggregation point where all battery string energy converges; it is the last line of defence before a fault propagates to the PCS or back into healthy battery racks.

DESCRIPTION

The DC Combiner & Distribution Panel is a floor-standing, IP54-rated steel cabinet (approx. 600 × 400 × 1800 mm) located on the short-end wall of the container adjacent to the PCS inverter. It aggregates the DC outputs of the 4–5 LFP battery racks onto a common tinned-copper busbar, then routes the combined DC bus via a single high-current cable pair to the PCS.

KEY PARTS (9)

- IP54 floor-standing steel enclosure, 600 × 400 × 1800 mm
- DC fuse holders with NH2 gPV fuses, 1500 V DC rated
- Main DC contactor, 1500 V DC / 1000 A
- DC arc-fault circuit interrupter (AFCI) module
- Copper busbar, tinned, 80 × 10 mm cross-section
- Hall-effect DC current transducer (LEM DHAB)
- Surge protection device (SPD) Type II
- Insulation monitoring device (IMD)
- Auxiliary 24 V DC control terminal block

FAILURE MODES (4)

- Fuse nuisance trip or coordination mis-match causing loss of a battery string
- DC contactor weld-closed failure preventing emergency isolation
- Insulation resistance degradation from moisture ingress or cable chafing
- Busbar hot-spot due to loose torque on bolted joints

OPEN QUESTIONS (3)

- Final number of rack feeders (4 vs 5) pending battery module/rack vendor selection
- Whether the PCS vendor requires integrated pre-charge circuitry inside the combiner
- DNO/insurer preference on grounded vs ungrounded (IT) DC topology

Estimated cost £10,890

SECTION JUDGEMENT — COMPOSITE SCORE: 8.2/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.	<i>I, Gemini, will cross-check mass totals automatically against transport limits.</i>
Visual Clarity	6/10	Good flow, but images with extreme aspect ratios create jarring layout shifts.	<i>I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.</i>
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.	<i>I, Gemini, will apply a readability filter to ensure consistent noun usage.</i>
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.	<i>I, Gemini, will map failure modes to the specific components they affect.</i>
Actionability	8/10	Reviews and risk matrices attached to modules provide direct mitigations.	<i>I, Gemini, will export risk mitigations to an external issue tracker.</i>

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Bidirectional PCS Inverter

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Converts bidirectionally between DC battery bus and 400 V AC at 500 kW for charge/discharge operation.

WHY IT MATTERS

The PCS is the single point of energy conversion between the battery and the grid and therefore sets the system's round-trip efficiency, grid compliance (G99 / IEEE 1547), response time for peak-shaving dispatch, and fault ride-through behaviour.

DESCRIPTION

The PCS is a floor-standing, cabinet-style bidirectional voltage-source converter rated at 500 kW continuous, converting between the 600–1,500 V DC battery bus and 400 V, 50 Hz, 3-phase AC. It is built around a 3-level IGBT bridge with an LCL output filter, DC pre-charge circuit, AC contactor, and LCL-side EMI filtering, all housed in an IP54 steel enclosure approximately 800 × 800 × 2,000 mm.

KEY PARTS (8)

- 500 kW bidirectional PCS unit
- IGBT power module stack (3-level NPC or T-type topology)
- LCL output filter with copper-wound inductors
- DC-side pre-charge circuit with contactor
- AC output contactor / motorised isolator, 1,000 A
- Digital control card with DSP + FPGA
- Internal forced-air cooling fans
- Integrated auxiliary 24 V DC supply and insulation-monitoring device (IMD)

FAILURE MODES (4)

- IGBT module failure due to thermal cycling or overcurrent
- DC-link capacitor degradation (ESR rise) leading to ripple overheating
- Output LCL filter inductor insulation breakdown
- Cooling fan or air-filter blockage causing thermal derate

OPEN QUESTIONS (3)

- Final DNO-agreed grid code settings (G99 Type A vs Type B)
- Whether a grid-forming (black-start / islanded) capability is required
- Confirmation of DC operating window

Estimated cost £113,400

SECTION JUDGEMENT — COMPOSITE SCORE: 8.2/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.	<i>I, Gemini, will cross-check mass totals automatically against transport limits.</i>
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Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.	<i>I, Gemini, will apply a readability filter to ensure consistent noun usage.</i>
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.	<i>I, Gemini, will map failure modes to the specific components they affect.</i>
Actionability	8/10	Reviews and risk matrices attached to modules provide direct mitigations.	<i>I, Gemini, will export risk mitigations to an external issue tracker.</i>

What is good:

What is bad:

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3.6 · AC Switchgear & Step-up Transformer

ADB

3200.0 kg / budget 3500.0 kg · 16 wk lead · Specialist judgement

AC Switchgear & Step-up Transformer

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Provides AC protection, metering, G99-compliant grid interface relays, and step-up from LV to MV for the DNO grid connection.

WHY IT MATTERS

This module is the legal and electrical interface between the BESS and the UK distribution network. Without a type-tested G99-compliant protection scheme and correctly-rated transformer, the DNO will not issue connection approval, rendering the entire product unsaleable.

DESCRIPTION

This module comprises the low-voltage AC switchgear cubicle located inside the container adjacent to the PCS, and an externally-skidded step-up distribution transformer sited alongside the container. The LV switchgear is a floor-mounted IP54 steel cubicle (powder-coated RAL 7035) housing the main 800 A MCCB, G99 protection relay, revenue meter, SPDs, auxiliary transformer feeder breaker, and terminal compartments for PCS-side and transformer-side cables.

KEY PARTS (8)

- LV AC switchgear cubicle, 400 V / 3-phase / 50 Hz, 800 × 600 × 2,000 mm, IP54
- G99-compliant protection relay
- Revenue-grade bidirectional energy meter, Class 0.5S
- Step-up distribution transformer, 630 kVA, 400 V Dyn11 ' 11 kV
- Auxiliary services transformer, 10 kVA, 400 V ' 230 V
- Surge protection device (SPD), Type 1+2
- LV cabling: 4-core 300 mm² XLPE/SWA/LSZH
- Earthing bar, 50 × 10 mm copper

FAILURE MODES (5)

- Protection relay nuisance tripping on grid disturbances
- Transformer winding insulation breakdown from repeated thermal cycling
- MCCB fails to clear a downstream fault within required time
- SPD degradation after lightning/surge events
- Auxiliary transformer overload during simultaneous HVAC + fire-pump operation

OPEN QUESTIONS (3)

- Final DNO connection voltage (11 kV vs 33 kV)
- Whether DNO will accept type-tested relay settings or require site-specific protection study
- Transformer siting: single external skid vs segregated internal compartment

Estimated cost £65,330

SECTION JUDGEMENT — COMPOSITE SCORE: 8.2/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.	<i>I, Gemini, will cross-check mass totals automatically against transport limits.</i>
Visual Clarity	6/10	Good flow, but images with extreme aspect ratios create jarring layout shifts.	<i>I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.</i>
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HVAC Thermal Management System

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Ceiling-mounted chiller and ducting that removes heat from battery racks and power electronics to maintain internal temperature envelope.

WHY IT MATTERS

LFP cell life is exponentially sensitive to temperature: sustained operation above 35 degC can halve calendar life and breach the 10-year warranty envelope. The HVAC system is therefore the primary enabler of the warranty, round-trip efficiency, and safe operation.

DESCRIPTION

The HVAC thermal management system is a rooftop-mounted, direct-expansion (DX) packaged chiller arrangement sized to reject the ~25–30 kW continuous heat load generated by the LFP battery racks and PCS at 1C operation, with ~50% thermal margin giving 40–50 kW nominal cooling capacity. Two redundant rooftop units (N+1) mount on reinforced curbs over roof penetrations.

KEY PARTS (9)

- Rooftop packaged DX chiller unit, 45 kW nominal cooling capacity
- Redundant secondary chiller unit, 45 kW (N+1 redundancy)
- Galvanised steel supply air ducting
- Return air plenum with aspirating intake grilles
- Condensate drain system
- Airflow and temperature sensors
- MERV-8 pre-filter and MERV-13 primary filter bank
- HVAC local controller
- Roof penetration curb kits

FAILURE MODES (4)

- Refrigerant leak (microcracks at brazed joints) causing loss of cooling capacity
- Condensate drain blockage or freeze-up leading to water ingress
- Filter clogging increasing fan static pressure
- Compressor failure or contactor welding in a single unit

OPEN QUESTIONS (3)

- Final confirmed heat load from selected cell vendor's UL 9540A data
- Whether UK DNO site ambient design temperature should be 35 degC or 40 degC
- Refrigerant selection lock-in: R454B vs R410A

Estimated cost £67,120

SECTION JUDGEMENT — COMPOSITE SCORE: 8.2/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.	<i>I, Gemini, will cross-check mass totals automatically against transport limits.</i>
Visual Clarity	6/10	Good flow, but images with extreme aspect ratios create jarring layout shifts.	<i>I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.</i>
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.	<i>I, Gemini, will apply a readability filter to ensure consistent noun usage.</i>
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.	<i>I, Gemini, will map failure modes to the specific components they affect.</i>
Actionability	8/10	Reviews and risk matrices attached to modules provide direct mitigations.	<i>I, Gemini, will export risk mitigations to an external issue tracker.</i>

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images push text down awkwardly if the aspect ratio is extreme.

320.0 kg / budget 350.0 kg · 10 wk lead · Specialist judgement

Fire Detection & Clean-Agent Suppression

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

NFPA 855 compliant aspirating smoke/off-gas detection, control panel, clean-agent cylinder, and ceiling nozzles for early-warning and suppression of battery fires.

WHY IT MATTERS

Lithium-ion battery fires are the single largest safety, insurance, and permitting risk for containerised BESS. Without NFPA 855 / UL 9540 compliant detection and suppression, the product cannot be insured, cannot obtain UK fire authority and DNO sign-off, and cannot be deployed on commercial sites.

DESCRIPTION

This module provides a two-stage fire safety system for the container: (1) very-early warning via aspirating smoke detection and lithium-ion off-gas sensing (H2, CO, electrolyte VOCs) that triggers alarms and shuts down the PCS/contactors before a thermal runaway propagates, and (2) clean-agent total-flooding suppression using a FK-5-1-12 (Novec 1230) cylinder and ceiling-mounted nozzle network.

KEY PARTS (9)

- Clean-agent cylinder, Novec 1230 (FK-5-1-12)
- Fire suppression control panel
- VESDA-style aspirating smoke detector
- Off-gas / H2 / CO / LEL gas detector array
- Ceiling discharge nozzles
- Schedule 40 seamless steel agent distribution piping
- Manual pull station and abort switch
- Deflagration vent panel on container roof
- Strobe/sonalert alarm devices

FAILURE MODES (4)

- False discharge of clean agent due to single-sensor fault
- Failure to detect early off-gas because sampling tubes become blocked
- Clean-agent flooding fails to prevent re-ignition of a battery in thermal runaway
- Deflagration vent fails to open at design pressure

OPEN QUESTIONS (3)

- Final agent quantity and nozzle layout pending CFD / hydraulic calculation
- Whether the insurer/DNO will accept clean agent alone or mandate a supplementary water-mist system
- Calibration and replacement interval for off-gas sensors

Estimated cost £28,430

SECTION JUDGEMENT — COMPOSITE SCORE: 8.2/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.	<i>I, Gemini, will cross-check mass totals automatically against transport limits.</i>
Visual Clarity	6/10	Good flow, but images with extreme aspect ratios create jarring layout shifts.	<i>I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.</i>
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.	<i>I, Gemini, will apply a readability filter to ensure consistent noun usage.</i>
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.	<i>I, Gemini, will map failure modes to the specific components they affect.</i>
Actionability	8/10	Reviews and risk matrices attached to modules provide direct mitigations.	<i>I, Gemini, will export risk mitigations to an external issue tracker.</i>

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images push text down awkwardly if the aspect ratio is extreme.

BMS / SCADA Control Panel

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Wall-mounted master BMS, EMS/SCADA HMI, and communications gateway coordinating all subsystems and interfacing to the site operator and grid.

WHY IT MATTERS

This module is the "brain" of the BESS: it enforces safe operating limits (SOC, temperature, voltage, current), coordinates charge/discharge dispatch against the grid tariff signal, and is the master safety interlock.

DESCRIPTION

The BMS/SCADA control panel is a wall-mounted IP54 steel cubicle (~800 × 300 × 1800 mm) located on the inner wall adjacent to the personnel service door, providing the single point of supervisory control, monitoring, and external communications for the entire BESS.

KEY PARTS (8)

- Master BMS controller (system-level BAU)
- EMS/SCADA industrial PC with 15" touchscreen HMI
- Managed industrial Ethernet switch
- 4G/LTE + fibre cellular gateway with VPN
- Auxiliary 24 V DC power supply with integrated UPS battery backup
- IP54 wall-mount steel enclosure
- Protection relays and I/O module
- Surge protection devices (Type 2 SPD)

FAILURE MODES (4)

- Loss of 24 V DC auxiliary power causing silent loss of monitoring
- Comms link failure between master BMS and rack BMUs
- Cybersecurity compromise of the 4G/fibre gateway
- HMI touchscreen or industrial PC failure due to thermal cycling

OPEN QUESTIONS (3)

- Final SCADA protocol stack required by the end customer
- Whether the DNO requires a hardwired G99 interface trip relay
- Cybersecurity certification target (IEC 62443-4-2 level)

Estimated cost £15,130

SECTION JUDGEMENT — COMPOSITE SCORE: 8.2/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.	<i>I, Gemini, will cross-check mass totals automatically against transport limits.</i>
Visual Clarity	6/10	Good flow, but images with extreme aspect ratios create jarring layout shifts.	<i>I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.</i>
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.	<i>I, Gemini, will apply a readability filter to ensure consistent noun usage.</i>
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.	<i>I, Gemini, will map failure modes to the specific components they affect.</i>
Actionability	8/10	Reviews and risk matrices attached to modules provide direct mitigations.	<i>I, Gemini, will export risk mitigations to an external issue tracker.</i>

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images push text down awkwardly if the aspect ratio is extreme.

4. BOM master (18 rows)

ADB

BOM derived from the module decomposition, expanded into typed part rows. Each row links back to its source module.

PART #	NAME	MODULE	TYPE	PROCESS / MATERIAL	MASS	COST
ENC-40FT-HC	40ft ISO High-Cube Steel Container — Modified ISO 1AAA container with reinforced floor.	Container Enclosure & Structure	buy	—	3800.00 kg	£12,000
ENC-INS-RW	Rockwool Fire Insulation — 50mm thermal and fire-rated lining.	Container Enclosure & Structure	buy	—	450.00 kg	£2,500
BAT-LFP-280	CATL 280Ah LFP Prismatic Cell — High cycle life LFP cell.	Port-side LFP Battery Rack Array	buy	—	5.50 kg	£120
BAT-RACK-FRM	Rack Steel Frame — Welded and powder-coated steel frame.	Port-side LFP Battery Rack Array	make	Laser & Brake · S275 Steel	150.00 kg	£800
BAT-LFP-280-SB	CATL 280Ah LFP Prismatic Cell — High cycle life LFP cell.	Starboard-side LFP Battery Rack Array	buy	—	5.50 kg	£120
BAT-RACK-FRM-SB	Rack Steel Frame — Welded and powder-coated steel frame.	Starboard-side LFP Battery Rack Array	make	Laser & Brake · S275 Steel	150.00 kg	£800
DCC-BUS-80	Tinned Copper Busbar 80x10mm — Main aggregation busbar.	DC Combiner & Distribution Panel	make	Machining · Copper	45.00 kg	£450
DCC-FUSE-1500V	15kV gPV Fuse 1500V — String isolation fuse.	DC Combiner & Distribution Panel	buy	—	2.10 kg	£85
INV-SIC-1MW	SMA 500kW PCS Unit — Grid-forming bidirectional inverter.	Bidirectional PCS Inverter	buy	—	1100.00 kg	£85,000
INV-LCL-FLT	LCL Output Filter — Harmonic suppression filter.	Bidirectional PCS Inverter	buy	—	85.00 kg	£3,200
XFM-630KVA	630kVA Step-up Transformer — 400V to 11kV distribution transformer.	AC Switchgear & Step-up Transformer	buy	—	2400.00 kg	£22,000
SWG-MCCB-800	800A MCCB — Main LV AC breaker.	AC Switchgear & Step-up Transformer	buy	—	12.00 kg	£1,500
HVAC-40KW-EXT	45kW Packaged Chiller — Rooftop DX chiller unit.	HVAC Thermal Management System	buy	—	280.00 kg	£18,000
HVAC-DCT-GALV	Galvanised Ducting — Internal air distribution.	HVAC Thermal Management System	make	Bending · Galvanised Steel	40.00 kg	£600
FIRE-NOVEC-CY	Novec 1230 Cylinder — Clean-agent suppression tank.	Fire Detection & Clean-Agent Suppression	buy	—	150.00 kg	£5,400
FIRE-SENS-H2	H2 Off-gas Sensor — Early warning off-gas detector.	Fire Detection & Clean-Agent Suppression	buy	—	0.50 kg	£800
BMS-MST-CTRL			buy	—	2.50 kg	£3,500

	Master BMS Controller — System-level battery management unit.	BMS / SCADA Control Panel			
BMS-HMI-15	15" Touchscreen HMI — Local operator interface.	BMS / SCADA Control Panel	buy	—	4.20 kg £1,200

Total cost	£158,075
Total mass	8682.30 kg
Buy / make	14 buy · 4 make

SECTION JUDGEMENT — COMPOSITE SCORE: 4.8/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	8/10	Part counts, masses, and costs mathematically aggregate correctly to system totals.	<i>I, Gemini, will add unit price validation against historical cost databases.</i>
Visual Clarity	3/10	A flat, unstyled list is illegible and causes severe cognitive overload.	<i>I, Gemini, will group the BOM hierarchically by Module/Sub-assembly. Right-align all numerical values so decimals align cleanly. Add light zebra striping for tracking.</i>
Linguistic Precision	6/10	Descriptions are accurate but lack standardized nomenclature.	<i>I, Gemini, will enforce a standardized naming convention (e.g., Noun, Adjective, Rating).</i>
Logical Coherence	4/10	Lacks structural sub-assembly grouping, failing to reflect the physical build hierarchy.	<i>I, Gemini, will infer a tree structure based on sourceModuleName.</i>
Actionability	3/10	Hard to use for procurement without clear module-level rollups.	<i>I, Gemini, will add supplier IDs directly to the BOM rows.</i>

What is good:	Comprehensive listing of components with correct mass and cost aggregations.
What is bad:	A raw list of parts is mind-numbing to read. BOM is a flat list without hierarchy.

5. Cost waterfall

ADB

Cost estimates derived from the module decomposition and the project's cost-estimate data. Per-module figures roll up to the unit cost shown on the cover page.

UNIT COST (ALL-IN)

£494,029

CEILING (BRIEF)

£180,000

HEADROOM

£314,029OVER

Per-module roll-up

MODULE	COST	% OF UNIT
Container Enclosure & Structure	£41,740	8.4%
Lithium Iron Phosphate Battery Rack Assembly	£26,703	5.4%
Bi-directional Grid-Forming Power Conversion System	£287,300	58.2%
Liquid Cooling Skid	£87,370	17.7%
Battery Management System	£10,660	2.2%
Fire Detection and Suppression System	£23,235	4.7%
Auxiliary Power and Low-Voltage Distribution	£17,021	3.4%
Controls and Supervisory Cabinet	£13,300	2.7%

SECTION JUDGEMENT — COMPOSITE SCORE: 4.4/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	8/10	Cost aggregates are mathematically sound and reflect the BOM.	I, Gemini, will add variance checks against the stated brief ceiling.
Visual Clarity	2/10	Displays as a flat accounting table with zero visual indication of cost drivers.	I, Gemini, will generate a real waterfall chart server-side (via QuickChart/Puppeteer), return as base64, and inject as an <Image> above the table.
Linguistic Precision	6/10	Standard financial terminology, but lacks contextual explanations.	I, Gemini, will add brief narrative summaries explaining major cost deviations.
Logical Coherence	4/10	Shows the numbers but fails to tell the story of the budget overrun.	I, Gemini, will explicitly link overrun categories to the module decisions that caused them.
Actionability	2/10	Does not visually highlight the areas requiring the most urgent cost engineering.	I, Gemini, will sort the table by cost magnitude and add percentage-of-total columns.

- What is good:

Clear aggregate vs per-module breakdown with headroom analysis.
- What is bad:

It reads as accounting output rather than a waterfall chart. Cost Waterfall lacks a real chart.

6. Risks register

FMEA-style risk matrix per module. Each row is rated severity (Negligible / Minor / Moderate / Major / Catastrophic) × likelihood (Rare / Unlikely / Possible / Likely / Frequent). Rating bands: low (1–3), medium (4–8), high (9–15), critical (16–25). Residual rating shows the band after the listed mitigation lands.

Container Enclosure & Structure

RISK MATRIX (2)

RM-1: Deflagration vent panels fail to open at design pressure during thermal runaway Cause: Vent panel hinge corrosion seizure Consequence: Container shell ruptures explosively Existing controls: Four NFPA 68 / EN 14797 certified vent panels Mitigation: Specify vent panels with stainless steel hinges Owner: Battery safety lead Residual: Major × Rare — medium (4)	Catastrophic × Unlikely — high (10)
RM-2: Corrosion perforation at sealant joints, roof penetrations, and gland plate Cause: Long-term degradation of polyurethane sealant Consequence: Water ingress causes insulation breakdown Existing controls: ISO 12944 C5-M paint system Mitigation: Specify EN 60529 IP66 rating at all penetrations Owner: Mechanical lead Residual: Moderate × Unlikely — medium (6)	Major × Possible — high (12)

OPEN QUESTIONS (3)

- Whether deflagration venting via roof panels or forced mechanical ventilation is preferred
- Final floor-loading spec: is 3,000 kg/m^2 sufficient
- Whether the step-up transformer is internal or external

Port-side LFP Battery Rack Array

RISK MATRIX (2)

RM-1: Single-cell thermal runaway propagating to adjacent cells and modules Cause: Internal cell defect or overcharge Consequence: Catastrophic loss of container, possible deflagration event Existing controls: Lithium iron phosphate chemistry, per-cell voltage monitoring Mitigation: Specify grade A cells with full traceability and matrix testing to UL 9540A Owner: Battery safety lead Residual: Major × Rare — medium (4)	Catastrophic × Unlikely — high (10)
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RM-2: Coolant leak from cold-plate fitting or hose

Major × Possible — high (12)

Cause: Vibration-induced loosening of dripless couplings

Consequence: Isolation fault, ground fault current, possible arc event

Existing controls: Dripless quick-disconnect couplings, drip trays

Mitigation: Add redundant leak-detection ribbon at base of every rack

Owner: Mechanical lead

Residual: Moderate × Unlikely — medium (6)

OPEN QUESTIONS (3)

- Final cell/module vendor selection (CATL, EVE, Gotion)
- Whether rack-level liquid cooling plates are required instead of forced-air
- DC string voltage target (1000 V vs 1500 V) pending final PCS selection

Starboard-side LFP Battery Rack Array

RISK MATRIX (2)

RM-1: Single-cell thermal runaway propagating to adjacent cells and modules

Catastrophic × Unlikely — high (10)

Cause: Internal cell defect or overcharge

Consequence: Catastrophic loss of container, possible deflagration event

Existing controls: Lithium iron phosphate chemistry, per-cell voltage monitoring

Mitigation: Specify grade A cells with full traceability and matrix testing to UL 9540A

Owner: Battery safety lead

Residual: Major × Rare — medium (4)

RM-2: Coolant leak from cold-plate fitting or hose

Major × Possible — high (12)

Cause: Vibration-induced loosening of dripless couplings

Consequence: Isolation fault, ground fault current, possible arc event

Existing controls: Dripless quick-disconnect couplings, drip trays

Mitigation: Add redundant leak-detection ribbon at base of every rack

Owner: Mechanical lead

Residual: Moderate × Unlikely — medium (6)

OPEN QUESTIONS (3)

- Final cell/module vendor selection (CATL, EVE, Gotion)
- Whether rack-level liquid cooling plates are required instead of forced-air
- DC string voltage target (1000 V vs 1500 V) pending final PCS selection

DC Combiner & Distribution Panel

RISK MATRIX (1)

RM-1: DC contactor weld-closed failure

Catastrophic × Unlikely — high (10)

Cause: Switching under heavy fault current

Consequence: Inability to isolate battery racks during emergency

Existing controls: Upstream NH2 gPV fuses

Mitigation: Size contactor for worst-case short circuit make/break capacity

Owner: Electrical lead

Residual: Moderate × Rare — low (3)

OPEN QUESTIONS (3)

- Final number of rack feeders (4 vs 5) pending battery module/rack vendor selection
- Whether the PCS vendor requires integrated pre-charge circuitry inside the combiner
- DNO/insurer preference on grounded vs ungrounded (IT) DC topology

Bidirectional PCS Inverter

RISK MATRIX (2)

RM-1: Silicon carbide power module short-circuit failure causing direct current bus fault

Catastrophic × Unlikely — high (10)

Cause: Power module die failure, gate driver malfunction

Consequence: Internal arc flash, cabinet damage, possible ignition source

Existing controls: Direct current input fuses and motorised circuit breaker

Mitigation: Specify silicon carbide modules with proven field reliability data

Owner: Electrical lead

Residual: Moderate × Rare — low (3)

RM-2: Grid-forming control loop instability under weak-grid or islanded conditions

Major × Possible — high (12)

Cause: Insufficient damping in virtual synchronous machine controller

Consequence: Voltage or frequency excursions tripping G99 protection

Existing controls: Virtual synchronous machine controller with configurable inertia

Mitigation: Run hardware-in-the-loop testing across the expected range of site short-circuit ratios

Owner: Controls lead

Residual: Moderate × Rare — low (3)

OPEN QUESTIONS (3)

- Final DNO-agreed grid code settings (G99 Type A vs Type B)
- Whether a grid-forming (black-start / islanded) capability is required
- Confirmation of DC operating window

AC Switchgear & Step-up Transformer

RISK MATRIX (1)

RM-1: Transformer winding insulation breakdown

Catastrophic × Unlikely — high (10)

Cause: Repeated thermal cycling and harmonic heating

Consequence: Internal short circuit, catastrophic failure

Existing controls: Oil temperature monitoring, Buchholz relay

Mitigation: Specify cast-resin dry type transformer instead of oil-filled

Owner: Electrical lead

Residual: Major × Rare — medium (4)

OPEN QUESTIONS (3)

- Final DNO connection voltage (11 kV vs 33 kV)
- Whether DNO will accept type-tested relay settings or require site-specific protection study
- Transformer siting: single external skid vs segregated internal compartment

HVAC Thermal Management System

RISK MATRIX (1)

RM-1: Refrigerant leak inside container

Major × Possible — high (12)

Cause: Vibration-induced microcracks at brazed joints

Consequence: Loss of cooling leading to battery thermal derating

Existing controls: Factory brazing quality control

Mitigation: Implement vibration isolation mounts for compressors

Owner: Mechanical lead

Residual: Moderate × Unlikely — medium (6)

OPEN QUESTIONS (3)

- Final confirmed heat load from selected cell vendor's UL 9540A data
- Whether UK DNO site ambient design temperature should be 35 degC or 40 degC
- Refrigerant selection lock-in: R454B vs R410A

Fire Detection & Clean-Agent Suppression

RISK MATRIX (1)

RM-1: False discharge of Novec 1230

Moderate × Possible — high (9)

Cause: Single-sensor fault or dust ingress

Consequence: System shutdown and expensive agent refill

Existing controls: Cross-zoned detection requirement

Mitigation: Require double-knock coincidence from both smoke and off-gas sensors

Owner: Safety lead

Residual: Minor × Unlikely — medium (4)

OPEN QUESTIONS (3)

- Final agent quantity and nozzle layout pending CFD / hydraulic calculation
- Whether the insurer/DNO will accept clean agent alone or mandate a supplementary water-mist system
- Calibration and replacement interval for off-gas sensors

BMS / SCADA Control Panel

RISK MATRIX (1)

RM-1: Cybersecurity compromise of 4G gateway

Cause: Unpatched firmware vulnerability

Consequence: Malicious remote operation of BESS

Existing controls: VPN tunnel to customer network

Mitigation: Implement strict firewall rules and regular penetration testing

Owner: Software lead

Residual: Moderate × Rare — low (3)

Catastrophic × Unlikely — high (10)

OPEN QUESTIONS (3)

- Final SCADA protocol stack required by the end customer
- Whether the DNO requires a hardwired G99 interface trip relay
- Cybersecurity certification target (IEC 62443-4-2 level)

SECTION JUDGEMENT — COMPOSITE SCORE: 7.8/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	9/10	Identified failure modes (e.g. vent panel seizure) reflect actual industry lessons learned.	I, Gemini, will cross-reference identified risks with historical incident databases.
Visual Clarity	7/10	Left-border severity coloring makes scanning highly intuitive.	I, Gemini, will group risks by likelihood and severity into a standard 5x5 heatmap.
Linguistic Precision	8/10	Risk terminology (cause, consequence, mitigation) is well applied.	I, Gemini, will enforce stricter separation between hazard and consequence.
Logical Coherence	8/10	Links logically from identified hazard to mitigation and residual risk.	I, Gemini, will ensure mitigations directly address the root causes, not just consequences.
Actionability	7/10	Directly assigns ownership and clear mitigation steps, but hard to sort.	I, Gemini, will sort strictly by RPN (Severity × Likelihood) descending so critical risks always render first.

What is good:

The vertical card format with dynamic left-border colors based on severity is excellent for scannability.

What is bad:

We need a way to filter or surface only the critical risks at the very top of the document.

7. Supplier shortlist (5)

'LLM | DEEPSEEK-V4-FLASH

Shortlist built by scoring each supplier in the directory against each module's declared process and material.

SUPPLIER	SCORE	LOCATION	MODULES
CATL	98%	Ningde, China	Port-side LFP Battery Rack Array, Starboard-side LFP Battery Rack Array
SMA Solar Technology	92%	Niestetal, Germany	Bidirectional PCS Inverter
Carrier	89%	Palm Beach Gardens, USA	HVAC Thermal Management System
ABB	95%	Zurich, Switzerland	AC Switchgear & Step-up Transformer
Johnson Controls (Tyco Fire)	91%	Cork, Ireland	Fire Detection & Clean-Agent Suppression

Supplier details

CATL

<https://www.catl.com>

Contact: sales@catl.com

Ningde, China · founded 2011 · 110k employees · lead time 12-16 weeks · MOQ 1 MWh

Certifications: ISO 9001, ISO 14001, UL 1973, IEC 62619

CATL is the global leader in LFP cell manufacturing. Their 280Ah and 314Ah prismatic cells perfectly match the density and cycle life requirements for this 3.5MWh BESS design. They possess robust supply chains capable of delivering the requested 15-rack volume.

- Industry leading LFP cell technology
- Proven track record in grid-scale BESS
- UL 1973 certified cells

Supplies BOM rows: BAT-LFP-280, BAT-LFP-280-SB

Match score: 98%

Ramp role: Primary Cell Supplier

SMA Solar Technology

<https://www.sma.de>

Contact: bess-sales@sma.de

Niestetal, Germany · founded 1981 · 4k employees · lead time 20 weeks · MOQ 1 Unit

Certifications: ISO 9001, UL 1741

SMA provides highly reliable grid-scale string and central inverters. Their modular inverter blocks align well with the 40ft containerized layout, allowing for scalable power conversion and straightforward integration with the central EMS.

- Grid-forming capability
- Strong European service network

Supplies BOM rows: INV-SIC-1MW, INV-LCL-FLT

Match score: 92%

Ramp role: Inverter Supplier

Carrier

<https://www.carrier.com>

Contact: commercial@carrier.com

Palm Beach Gardens, USA · founded 1915 · 53k employees · lead time 8 weeks · MOQ None

Certifications: ISO 9001, CE Marking

Carrier offers industrial-grade HVAC units capable of the strict 25degC baseline required by the LFP cells. Their external wall-mounted units fit within the ISO 40ft container footprint constraints.

- High reliability HVAC systems
- Global support network

Supplies BOM rows: HVAC-40KW-EXT

Match score: 89%

Ramp role: HVAC Supplier

ABB

<https://global.abb/group/en>

Contact: contact.center@ch.abb.com

Zurich, Switzerland · founded 1988 · 105k employees · lead time 14 weeks · MOQ 1 Unit

Certifications: ISO 9001, IEC 61439

ABB's LV switchgear and distribution transformers are perfectly suited for this grid-tied application, providing excellent reliability and G99 compliant protection.

- Extensive LV/MV portfolio
- G99 compliant protection relays

Supplies BOM rows: XFM-630KVA, SWG-MCCB-800

Match score: 95%

Ramp role: Switchgear & Transformer Supplier

Johnson Controls (Tyco Fire)

<https://www.johnsoncontrols.com>

Contact: fire.suppression@jci.com

Cork, Ireland · founded 1885 · 100k employees · lead time 6 weeks · MOQ 1 System

Certifications: NFPA 855, UL, FM Approved

Tyco provides state-of-the-art clean-agent fire suppression systems incorporating Novec 1230 and highly sensitive off-gas detection crucial for BESS safety.

- NFPA 855 expertise
- Comprehensive clean-agent portfolio

Supplies BOM rows: FIRE-NOVEC-CYL, FIRE-SENS-H2

Match score: 91%

Ramp role: Fire Suppression Supplier

SECTION JUDGEMENT — COMPOSITE SCORE: 6.2/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	8/10	Supplier metrics and capabilities match reality for CATL, SMA, and Carrier.	<i>I, Gemini, will pull live certifications from public ISO/UL registries.</i>
Visual Clarity	4/10	Renders like a flat database dump rather than a curated commercial shortlist.	<i>I, Gemini, will move to a Card-based UI. Use flexbox to make 2 columns of supplier cards, truncating the reason slightly for visual uniformity.</i>
Linguistic Precision	7/10	Project synthesis statements are well-written and justify the matches.	<i>I, Gemini, will ensure synthesis statements are under 50 words for brevity.</i>
Logical Coherence	7/10	Connects parts to suppliers logically, but the presentation flow is disjointed.	<i>I, Gemini, will group suppliers by their critical path impact (e.g. Longest Lead Time first).</i>
Actionability	5/10	Hard to quickly compare lead times and certifications side-by-side.	<i>I, Gemini, will add a comparative summary table before the detailed supplier cards.</i>

What is good: Scores and match reasons are highly actionable.

What is bad: It looks like a dumped directory rather than a curated shortlist.

8. Project audit log (2)

Actions recorded against this project — brief lock, ship, other auditable mutations.

- 25 Apr 2026, 05:42

Project created [Meta] Project initiated
- 25 Apr 2026, 18:30

Brief locked [Brief] Constraints and targets finalized

SECTION JUDGEMENT — COMPOSITE SCORE: 2.8/10

Dimension	Score	Critique & Rationale	Next-Round Code Fix (Gemini)
Factual Integrity	10/10	Immutable logs are perfectly accurate to the event timeline.	I, Gemini, will add cryptographic hashes to ensure log integrity.
Visual Clarity	1/10	Spans multiple pages with massive, redundant whitespace and huge fonts.	I, Gemini, will strip it down. Drop the font size to 6pt. Use a 3-column flexWrap grid to densely pack the logs like a machine readout.
Linguistic Precision	1/10	Robotic, repetitive system strings with no human-friendly summarisation.	I, Gemini, will pass raw logs through an LLM summarisation step to group repetitive actions.
Logical Coherence	1/10	No grouping or filtering; simply a chronological dump.	I, Gemini, will group logs by user or subsystem rather than pure chronology.
Actionability	1/10	Impossible to extract meaningful insights without painstaking reading.	I, Gemini, will highlight critical configuration changes in bold to distinguish them from read-only events.

- What is good:

Complete immutability tracking.
- What is bad:

Audit Log is a junk drawer taking up multiple pages of useless timestamps.